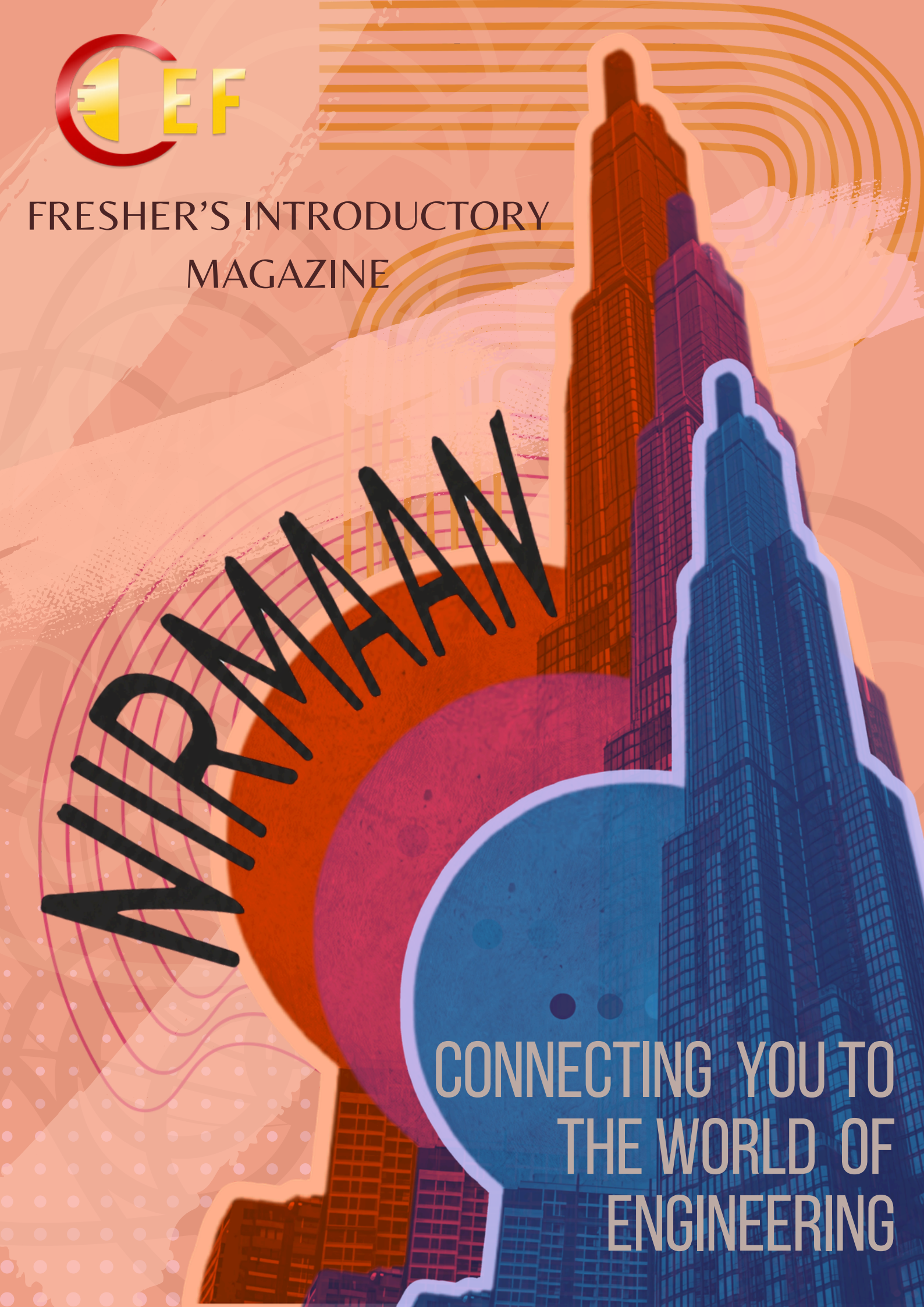
CEF

FRESHER'S INTRODUCTORY MAGAZINE

NIRMAAN



CONNECTING YOU TO THE WORLD OF ENGINEERING

PRIMARY

VOL. V

LETTER FROM THE DESIGN TEAM

Design is where imagination meets expression, and for us at the Civil Engineering Forum (CEF), working on Infravision has been an opportunity to translate the department's stories, ideas, and achievements into a visual language that resonates.

This edition of Infravision was a labor of creativity and collaboration. Our aim was to ensure that every page not only complements the content but also reflects the vibrant, innovative, and inclusive spirit of our civil engineering community.

I'd like to extend heartfelt appreciation to Srishti and Aditi, whose dedication, creativity, and attention to detail brought so much character to this magazine. Their work has been invaluable in shaping the look and feel of this edition, and it's been a pleasure working alongside such talented executives.

~Jyoti
Design Coordinator, CEF

EDITORS CONTENT TEAM

At the CEF, our goal has always been to create spaces that foster knowledge, creativity, and community spirit among students. Infravision , the official magazine of CEF, is a step in that direction, a platform to showcase the stories, innovations, and achievements that make our department truly special.

Through carefully curated articles, we've tried to bring to light research breakthroughs, technical projects, alumni insights, student experiences, and much more. Every piece has been written with the intent to inform, inspire, and initiate dialogue among peers.

As the Content Team Lead, I've had the privilege of working alongside an incredibly talented group of writers and thinkers, including Utkarsh, Aayush, Vishnu, Hanshie aur Samriddhi who poured time, thought, and effort into creating the content you're about to read. Each article is a result of deep discussions, shared curiosity, and a love for storytelling.

~Prakhar
Content and Publicity Coordinator, CEF

HOD'S NOTE

Dear Readers,

It is with pleasure that I welcome you to read through the latest edition of Nirmaan, the flagship publication of our Civil Engineering Forum in the Department at IIT Delhi. This magazine represents not just our academic endeavours, but also the collective spirit of innovation and excellence that defines our community.

Civil Engineering continues to be the cornerstone of societal development and progress. From designing earthquake-resistant structures to pioneering sustainable water management systems, from revolutionizing transportation networks to creating eco-friendly building materials - our discipline touches every aspect of human civilization. In this context, I am immensely proud to announce that our commitment to excellence has been recognized internationally, with our Department achieving the prestigious distinction of being ranked #1 in India and #39 globally in the QS World University Rankings by Subject 2024. This remarkable achievement reflects the tireless dedication of our faculty, students, and staff.

At IIT Delhi, we have cultivated an academic ecosystem that nurtures curiosity, critical thinking, and encourages interdisciplinary collaboration. Our department takes immense pride in its stateof-the-art facilities that rival the best in the world. Our robust network of industry partnerships and research collaborations further enriches the educational experience, offering students an exposure to real-world challenges and opportunities. The pages of Nirmaan offer you a window into this vibrant academic community - showcasing groundbreaking research initiatives, highlighting student achievements, featuring faculty contributions to the field, and documenting the myriad events that make our Department a hub of intellectual activity. I would like to extend my heartfelt appreciation to the dedicated members of the Civil Engineering Forum team who have invested countless hours in conceptualizing, curating, and crafting this comprehensive edition. As you peruse through this issue, I hope you will gain valuable insights into the dynamic world of Civil Engineering at IIT Delhi. Thank you for your enduring support and engagement with our academic community. We look forward to continuing this journey of excellence and innovation with you by our side.

With warm regards and sincere appreciation,

Vasant Matsagar

HOD

Dept. of Civil & Environmental Engineering ,IIT Delhi

GEN SEC'S NOTE

It is with immense pride and enthusiasm that I bring to you this edition of Nirmaan, the magazine of the Civil Engineering Forum (CEF), IIT Delhi.

For me, CEF has always been much more than a departmental society. It has been one of the most special parts of my journey at IIT Delhi- a place where I have learned, contributed, met some wonderful people and grown in ways that extend far beyond academics. I joined CEF as a team member with nothing more than a curiosity to understand how things worked behind the scenes, how a team comes together and how an idea is turned into something meaningful for students. To now have the opportunity to lead the same forum as its General Secretary is a journey I take immense pride in.

Over the years, I have also had the opportunity to witness CEF evolve. With every team, we have tried to build upon what came before us, to become more engaging, more accessible, and more connected to the students. From Extravaganza and departmental dinners to technical sessions, workshops, online competitions, Inter-IIT sports tournament, field trips, orientations, seminar series, informal interactions and guidance sessions, the forum has continuously tried to bring together the academic and non-academic sides of student life. And perhaps that is what makes CEF special.

Nirmaan is an extension of that spirit. This edition captures the many dimensions of our Civil Engineering community, our department, CEF's initiatives, student achievements, technical insights, and the diverse activities that shape life within the department. More than simply documenting what we have done, I hope these pages reflect the energy, ideas, and enthusiasm that keep our community moving forward.

Also, I am deeply grateful to our faculty advisors for their constant guidance and faith in us.And to every student, coordinator, volunteer, and contributor who has been a part of CEF-thank you. Your enthusiasm is what gives this forum life. Here's to CEF- to everything we have built, everything we continue to build, and everything that lies ahead.

Mehul Kumar
General Secretary
Civil Engineering Forum

PRESIDENT'S NOTE

As the President of the Civil Engineering Forum (CEF), I am thrilled to invite all of you to actively participate in our vibrant and diverse activities. At CEF, we aim to provide a refreshing platform that balances academic rigor with opportunities for personal and professional growth. From engaging seminars and career guidance sessions to fun-filled events like sports extravaganzas, jamming nights, and talent showcases, CEF has something for everyone.

These extracurricular activities are designed not only to enhance your skills and network but also to offer a welcome break from the intense academic schedule of IIT Delhi. By joining us, you will be part of a community that fosters creativity, camaraderie, and a shared passion for civil engineering. Let's come together, ease the pressure, and make your journey at IIT both fulfilling and memorable!

Sovik Das
President
CEF, IIT Delhi

VICE PRESIDENT'S NOTE

As the current Vice President of Civil Engineering Forum (CEF), I get to closely work with an incredible group of hardworking and talented students. This magazine reflects, not just the efforts of the current team, but also the dedication of everyone, who has contributed to CEF over the years.

CEF organizes a wide range of academic and non-academic activities throughout the year. These include seminars, guest lectures, site visits, departmental programmes, sport events, and networking opportunities. These events complement classroom learning and allow students to develop leadership, teamwork, and a shared passion for civil engineering. More importantly, they create friendships and connections with peers, faculty, and staff that stay with us long after we leave IIT Delhi.

I strongly encourage the students and faculty to be part of the CEF community and make the most of their journey here.

Allan Lambor Marbaniang
Vice President
CEF , IIT DELHI

EX PRESIDENT'S NOTE

I used to travel to the lakeside near my home during my early years. There, I regularly saw the synergistic harmonious dance between the blue waves and enchanting white birds. I used to enjoy this romance, as it used to fill me with energy and a pint of tranquillity. Years passed, and at dusk, I went to the same lakeside. However, there was no trace of the birds, and the waves were dark and heavy. Disheartened, I sat near a mango tree. Feeling the unknown but certain earth rotating underneath me and watching the sun disappear through the horizon, I wondered about the birds.
Suddenly, I remembered someone once saying: when we see the stars, we don't really know they are still there; they might as well have disappeared a long time ago. Straightaway, I realized, despite its occasional misfortunes, the present is all we have, and I stood up to my feet.

We're just a tiny ripple on an immense sea, What the future awaits for us is undoubtedly unknown, But while the clock is still ticking, We might as well enjoy the ride.

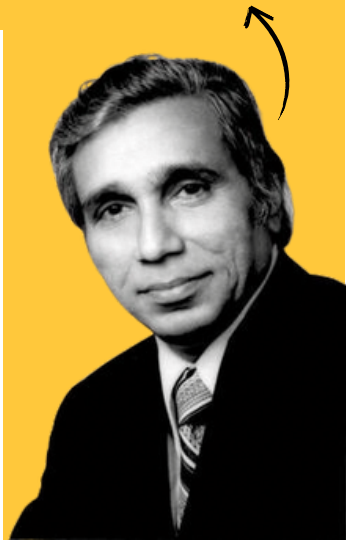
So, get out there and experience as much as you can. Participate in each unit of life with happiness and untiring commitment.

Deepanshu Shirole
EX PRESIDENT
CEF, IIT Delhi

FAZLUR RAHMAN KHAN

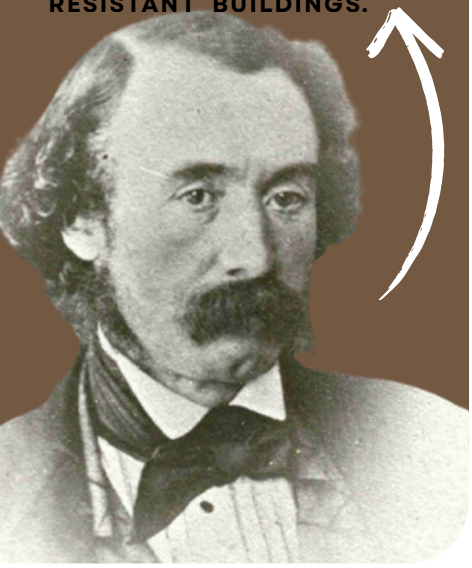
- BANGLADESHI-AMERICAN ARCHITECT AND STRUCTURAL ENGINEER
- PIONEERED CRITICAL STRUCTURAL SYSTEMS FOR SKYSCRAPERS
- RECOGNIZED AS ONE OF THE MOST INFLUENTIAL ENGINEERS OF THE 20TH CENTURY
- KNOWN AS THE "FATHER OF TUBULAR DESIGN FOR HIGH RISES" (ASCE)
- INNOVATOR IN COMPUTER-AIDED DESIGN FOR STRUCTURAL ENGINEERING
- NOTABLE WORKS

1. WILLIS TOWER – 2ND TALLEST TOWER IN THE UNITED STATES
2. JOHN HANCOCK CENTER – 100-STORY ICONIC STRUCTURE



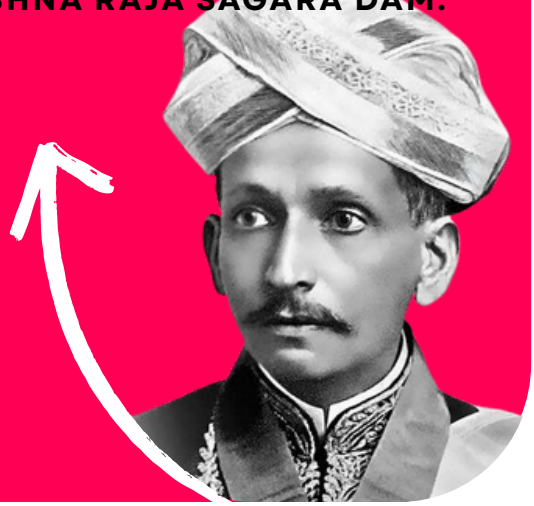
ROBERT MALLET

- DATE OF BIRTH TO DATE OF DEATH: JUNE 3, 1810 – NOVEMBER 5, 1881
- INTRODUCTION: AN IRISH CIVIL ENGINEER AND SEISMOLOGIST, RENOWNED FOR HIS WORK IN EARTHQUAKE STUDIES.
- CONTRIBUTIONS TO CIVIL ENGINEERING: A PIONEER IN SEISMOLOGY AND EARTHQUAKE ENGINEERING, HIS RESEARCH LAID THE FOUNDATION FOR UNDERSTANDING SEISMIC BEHAVIOR IN STRUCTURES AND CONTRIBUTED TO THE DESIGN OF EARTHQUAKE-RESISTANT BUILDINGS.



DR. MOKSHAGUNDAM VISVESVARAYA

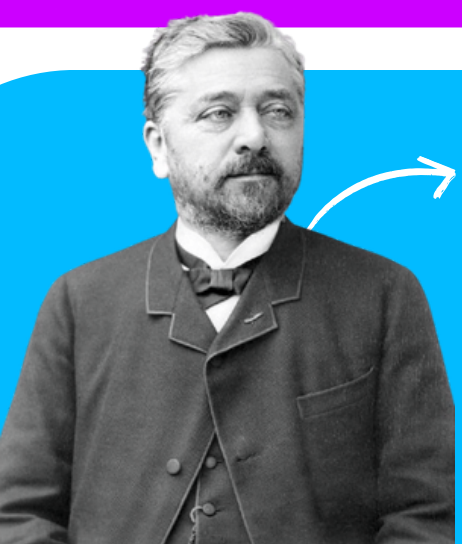
- DATE OF BIRTH TO DATE OF DEATH: SEPTEMBER 15, 1861 – APRIL 12, 1962
- INTRODUCTION: A RENOWNED INDIAN ENGINEER, STATESMAN, AND KEY FIGURE IN THE DEVELOPMENT OF MODERN KARNATAKA.
- CONTRIBUTIONS TO CIVIL ENGINEERING: A PIONEERING CIVIL ENGINEER RECOGNIZED FOR HIS WORK IN IRRIGATION AND DAM CONSTRUCTION, INCLUDING THE DESIGN AND CONSTRUCTION OF THE KRISHNA RAJA SAGARA DAM.



KARL TERZAGHI

- DATE OF BIRTH TO DATE OF DEATH: OCTOBER 2, 1883 – OCTOBER 25, 1963
- INTRODUCTION: AN AUSTRIAN-AMERICAN GEOTECHNICAL ENGINEER, WIDELY REGARDED AS THE "FATHER OF SOIL MECHANICS."
- CONTRIBUTIONS TO CIVIL ENGINEERING: MADE GROUNDBREAKING CONTRIBUTIONS TO SOIL MECHANICS AND FOUNDATION ENGINEERING, ESTABLISHING THEORIES AND PRINCIPLES ESSENTIAL FOR SAFE CONSTRUCTION ON VARIOUS SOIL TYPES.

The OGs



GUSTAV EIFFEL

- FRENCH CIVIL ENGINEER AND ARCHITECT
- KNOWN FOR THE GARABIT VIADUCT, EIFFEL TOWER, AND STATUE OF LIBERTY FRAMEWORK
- CONTRIBUTED SIGNIFICANTLY TO THE FRENCH RAILWAY NETWORK
- FOCUSED ON AERODYNAMICS AND METEOROLOGY AFTER RETIRING
- FOUNDED THE EIFFEL STEEL FIRM AND EXECUTED GLOBAL PROJECTS
- A PIONEER OF CIVIL ENGINEERING DURING THE INDUSTRIAL REVOLUTION
- FRANCE'S LEGACY

1. HOME TO ICONIC CIVIL ENGINEERING MARVELS LIKE THE EIFFEL TOWER AND MILLAU VIADUCT
2. A BIRTHPLACE OF RENOWNED ENGINEERS AND INNOVATIVE STRUCTURES

FLAGSHIP EVENTS @ CEF

FRESHER'S PARTY

WHAT'S THE BEST WAY TO BREAK THE ICE, MEET YOUR BATCHMATES, AND MAKE YOUR FIRST COLLEGE MEMORIES? WAIT FOR IT..... FRESHER'S PARTY(COULD HAVE NEVER GUESSED)! IMAGINE AN EVENING FILLED WITH MUSIC, GAMES, AND PERFORMANCES — PLUS THAT ELECTRIC BUZZ OF MEETING NEW FACES WHO WILL SOON BECOME LIFELONG FRIENDS (WHETHER YOU LIKE THEM OR NOT). IF YOU'RE AIMING FOR THE COVETED TITLE OF MR. OR MRS. FRESHER (OR FOR THE FOOD(MY KIND OF PEOPLE)), THIS IS YOUR MOMENT—DON'T EVEN THINK ABOUT MISSING IT!

DEPT DINNER

HAVE YOU EVER WONDERED HOW THE PROFESSORS ARE IN THE CIVIL ENGINEERING DEPARTMENT? WELL, NOW'S YOUR CHANCE TO MEET THEM LIKE NEVER BEFORE—WITHOUT THE STRESS OF ASSIGNMENTS HANGING OVER YOUR HEAD... AND MAYBE EVEN CATCH THEM CRACKING A JOKE (YES, THEY HAVE A SENSE OF HUMOUR)! THEY SAY FOOD BRINGS PEOPLE TOGETHER (PROBABLY BECAUSE IT'S HARD TO ARGUE WITH YOUR MOUTH FULL), SO WHY NOT A DINNER FOR THIS VERY CAUSE? IT'S NOT JUST A MEAL; IT'S WHERE BONDS ARE MADE, STORIES ARE TOLD, AND THE CIVIL ENGINEERING FAMILY EATS... I MEAN, MEETS!

ALUMNI TALKS

NO SCOPE IN CIVIL?CEF'S ALUMNI TALKS WILL SET YOU STRAIGHT! MEET THE SENIORS WHO WERE ONCE IN YOUR SHOES AND ARE NOW ROCKING THEIR CAREERS. AN EYE-OPENING EVENT THAT'LL CHANGE YOUR PERSPECTIVE, AND DON'T FORGET THE SNACKS! BECAUSE, LET'S FACE IT, DEEP THOUGHTS ARE WAY BETTER WITH A SAMOSA IN HAND!

CIVIL SEMINAR SERIES

ARE YOU READY TO EXPLORE GROUNDBREAKING IDEAS SHAPING THE FUTURE OF CIVIL ENGINEERING? THE CIVIL SEMINAR SERIES IS HERE TO IGNITE YOUR CURIOSITY AND CONNECT YOU WITH THE GLOBAL PIONEERS IN THE FIELD, WITH SPEAKERS FROM TOP UNIVERSITIES AROUND THE WORLD! WHETHER YOU'RE A BTECH, MTECH, OR PHD STUDENT, THIS SERIES OFFERS SOMETHING VALUABLE FOR ALL! ASK QUESTIONS, SHARE IDEAS, AND NETWORK WITH EXPERTS AND PEERS. PLUS, THERE'LL BE SNACKS! BECAUSE WHAT'S A SEMINAR WITHOUT FREE FOOD?

TRYST X CEF

WHAT DO YOU GET WHEN YOU MIX CIVIL ENGINEERING GENIUS WITH TRYST CREATIVITY? A DAY OF EPIC FUN AND WILD IDEAS! FROM SHAKY TOWERS TO DEBATES ON FLOATING CITIES, IT'S A PERFECT MIX OF 'CAN WE ACTUALLY BUILD THAT?' AND 'OH NO, IT'S COLLAPSING!' THE CEF X TRYST COLLAB CHALLENGES YOU TO RETHINK SPACES AND STRUCTURES WITH A SIDE OF HUMOUR AND SNACKS. SO COME JOIN US THIS YEAR — LET'S BUILD SOME STRUCTURALLY UNFORGETTABLE MEMORIES!

EXTRAVAGANZA

GET READY TO UNLEASH YOUR INNER ATHLETE AT EXTRAVAGANZA! CRICKET, TABLE TENNIS, BADMINTON, DODGEBALL —WHATEVER YOUR GAME, WE'VE GOT IT! SPORTS HAVE THIS MAGICAL WAY OF CONNECTING PEOPLE: THE BONDS LAST LONGER THAN OUR 'CHAMPION' MOMENTS. WELCOME TO EXTRAVAGANZA, WHERE WE PLAY LIKE IT'S THE FINALS... BUT REALLY, WE'RE JUST HERE FOR THE FUN, THE LAUGHS, AND MAYBE A TINY BIT OF GLORY!

MEME COMP

THE MEME COMPETITION IS HERE TO MIX LAUGHTER WITH LATE-NIGHT ASSIGNMENTS AND CREATIVITY WITH CAD FILES! AND LET'S BE HONEST, WHAT'S BETTER THAN GETTING THE BRAGGING RIGHTS OF THE MEME MASTER OF THE DEPARTMENT (AND MAYBE SOME ECAS AS WELL)?

CORE PLACEMENT GUIDANCE SESSIONS

THE CORE PLACEMENT GUIDANCE SESSION IS HERE TO HELP YOU CONFIDENTLY POWER THROUGH PLACEMENT SEASON. GET TO KNOW WHAT IS REQUIRED TO EXCEL FROM YOUR ALUMNI AND PEOPLE DIRECTLY INVOLVED IN THE INDUSTRY (AND NOT FROM THE SELF-PROCLAIMED CIVIL ENGINEERING 'EXPERTS' OF OTHER DEPARTMENTS).

CONCRETE for Civ

Structural Engineering

THE BACKBONE OF CIVILIZATION

EVER WATCHED WORKERS PLACING STEEL RODS WHILE BUILDING A HOUSE AND WONDERED WHO FIGURES OUT THEIR ARRANGEMENT AND NUMBER? THAT'S WHERE STRUCTURAL ENGINEERS STEP IN! FROM DESIGNING EARTHQUAKE-RESISTANT BUILDINGS TO CREATING AWE-INSPIRING STADIUMS, THEY'RE THE MASTERMINDS BEHIND IT ALL. DON'T WORRY—YOU'LL GET TO DIVE INTO THIS FASCINATING FIELD TOO WHEN YOU STUDY COURSES LIKE CVL 243. WITH A FUTURE FULL OF EXCITEMENT, YOUR WORK COULD LITERALLY HOLD THE WORLD TOGETHER—OR AT LEAST A SHOPPING MALL!

Geotechnical Engineering

DIGGING DEEP FOR BIG DREAMS

DID YOU KNOW THE TALLEST STRUCTURES ON EARTH HAVE FOUNDATIONS THAT GO AS DEEP AS 50 METERS? HOW DID WE FIGURE THAT OUT? WHO KNEW DIGGING IN THE DIRT COULD BE SO REWARDING! GEOTECHNICAL ENGINEERS STUDY SOIL AND ROCKS TO ENSURE YOUR DREAM SKYSCRAPER DOESN'T SINK LIKE A DEFLATED SOUFFLÉ. YOU'LL GET A GLIMPSE INTO THE WORLD BENEATH OUR FEET WITH CVL 121, AND IF YOU'VE EVER BEEN "GROUNDED" BY YOUR PARENTS, COURSES LIKE CVL 321 IS WHERE YOU'LL TRULY SHINE. WITH TUNNELING PROJECTS, METRO SYSTEMS, AND MEGA-DAMS ON THE RISE, THIS SPECIALIZATION IS ALL ABOUT GETTING DOWN TO EARTH—LITERALLY!

Transportation Engineering

KEEP CALM AND COMMUTE ON

EVER WONDERED HOW THE ROADS TO THE MOUNTAINS ARE MADE OR IF YOU'VE EVER CURSED TRAFFIC JAMS OR CELEBRATED A SMOOTH RIDE, THANK A TRANSPORTATION ENGINEER! THESE WIZARDS OF MOBILITY ENSURE OUR ROADS, RAILWAYS, AND TRANSIT SYSTEMS RUN LIKE CLOCKWORK. WITH THE BUZZ AROUND DRIVERLESS CARS AND SMART CITIES, THIS FIELD IS SPEEDING TOWARDS A FUTURE YOU'LL WANT A FRONT-ROW SEAT FOR.

ineers KINGDOM

Environmental Engineering

SAVING THE PLANET, ONE DROP AT A TIME

EVER STEPPED OUTSIDE CAMPUS AND WONDERED, "WHAT'S UP WITH THIS FOG?" SPOILER ALERT: IT'S NOT FOG—IT'S HUMANITY'S GIFT TO NATURE! ENTER ENVIRONMENTAL ENGINEERS, THE UNSUNG HEROES FIGHTING POLLUTION, CLIMATE CHANGE, AND WASTE. IMAGINE DESIGNING WATER SYSTEMS THAT MAKE DESERTS BLOOM OR CREATING AIR FILTERS THAT KEEP CITIES BREATHABLE—IT'S NOT JUST A DREAM; IT'LL SOON BE YOUR REALITY IN COURSES LIKE CVL 100. PLUS, YOU'LL GET TO CASUALLY DROP AT PARTIES THAT YOU'RE SAVING THE PLANET—AND ACTUALLY MEAN IT! TALK ABOUT A GREEN (AND UNDENIABLY COOL) CAREER PATH.

Construction Engineering

THE ULTIMATE PROJECT JUGGLER

THINK OF CONSTRUCTION ENGINEERS AS THE SUPERHEROES OF BUILDING PROJECTS. THEY'RE THERE FROM START TO FINISH, MAKING SURE EVERYTHING RUNS SMOOTHLY—LIKE A MOVIE DIRECTOR, BUT WITH CRANES AND CONCRETE. WHEN UNEXPECTED PROBLEMS POP UP (AND THEY ALWAYS DO), THEY'RE THE ONES KEEPING THINGS ON TRACK WITH BACKUP PLANS. IT'S ALL ABOUT BALANCING DESIGN, PLANNING, AND MANAGEMENT, WHILE KEEPING THE CHAOS IN CHECK! IF YOU FEEL LIKE OTHER CIVIL ENGINEERING DOMAINS AREN'T QUITE YOUR THING, BUT YOU HAVE AN INTEREST IN ECONOMICS, MANAGEMENT, OR PLANNING, CONSTRUCTION ENGINEERING COULD BE A GREAT FIT FOR YOU!

Water Resources Engineering

TURNING THE TIDE

WATER, WATER EVERYWHERE, THAT'S HOW OUR CAMPUS BECAME DURING MONSOON WHAT IF I TELL IT'S YOUR JOB TO MANAGE IT WISELY IN COURSES LIKE CVL282! FROM DESIGNING FLOOD-RESISTANT CITIES TO ENSURING CLEAN DRINKING WATER, WATER RESOURCE ENGINEERS ARE THE UNSUNG HEROES OF SUSTAINABILITY. WITH GLOBAL WATER CHALLENGES MAKING HEADLINES, THIS FIELD OFFERS A TIDE OF OPPORTUNITIES TO MAKE WAVES IN YOUR CAREER!

SURVIVAL GUIDE
FUNDAE

Brace yourself, the labs are coming

Lab hours can be a test of endurance, but they're also where the magic happens. Stay ahead by reading the lab manual beforehand, and bring essentials like a notebook, calculator, and snacks (because hunger is the real enemy). Oh, and remember—when the TA asks you to "observe," don't just nod; actually, take notes. Someday, those notes might save you.

Curiosity is your golden ticket

If you want professors to trust you with projects, show genuine interest. Attend lectures regularly (yes, just being present works wonders) and don't shy away from asking thoughtful questions—something like, "How does this principle work in real-life ?" When professors see your curiosity, they'll be more inclined to guide you toward meaningful opportunities.

Build a Connection

If you're consistently doing well in a course, don't be afraid to reach out to professors for a Letter of Recommendation (LoR). Professors appreciate students who show dedication and a genuine interest in their field-having a conversation about future aspirations or internships could open doors you didn't expect.

Snooze button = Loose button

Grade reductions don't always come from missed submissions or exams-they can be triggered by poor attendance. Professors take attendance seriously, so make sure you show up. Set alarms, plan ahead, and maybe even consider wearing lecture clothes to bed if you're running late. Trust me, those missed hours could cost you more than you think.

Smart friends are underrated blessings.

Every class has that one person who are perfectly organized notes and always asks the questions you didn't even know existed. Befriend them! Offer chai, share snacks, and if you borrow their notes, treat them like treasure-return them neatly, and don't you dare dog-ear the pages.

Seniors are walking, talking cheat codes

They've been through it all-messy labs, impossible deadlines, and professors' quirks. Ask them for advice on surviving tricky courses, borrowing resources, or even choosing the best electives. They'll also know which assignments need originality and which ones are more, let's say, flexible. Treat them with respect-they're your mentors in disguise.

PYQs: Your Survival Kit for Exams

If exams feel overwhelming, PYQs are your shortcut to understanding what's important. Start solving them early, compare patterns, and focus on frequently repeated concepts. Combine them with CEF resources or drive links from your batchmates, and you're halfway there.

Pro Tip: Solve them with friend to make it less of a grind.

TAs : Your Unsung Allies

Your Teaching Assistants are often the first to review your assignments or labs. Greet them, show interest in their feedback, and always submit your work on time. A little appreciation for their effort goes a long way-they might even guide you toward a better grade.

Exam Tips: Partial Marking is Life

Blank spaces on exam sheets? Big No. Write what you know, even if it's just a diagram or a rough calculation. Professors appreciate effort, and partial marks are better than a zero any day.

INTERN KE FUNDAE

YOU HAVE TWO WHOLE YEARS. BEFORE YOU SERIOUSLY START WORRYING ABOUT INTERNSHIPS. OUR HONEST ADVICE FROM CEF? DON'T TURN THESE TWO YEARS INTO A RACE. EXPLORE, EXPERIMENT, HAVE FUN, JOIN CLUBS, BUILD FRIENDSHIPS—AND ALONG THE WAY, SPEND SOME TIME FIGURING OUT WHAT ACTUALLY EXCITES YOU

INTERNSHIPS CAN TAKE YOU IN MANY DIRECTIONS. HERE ARE A FEW COMMON ONES TO HELP YOU UNDERSTAND WHAT MIGHT FIT YOU:

1.CORE

IF BUILDINGS, INFRASTRUCTURE AND ENGINEERING GENUINELY EXCITE YOU, CORE IS YOUR PLAYGROUND. EXPLORE CONSTRUCTION, STRUCTURAL, TRANSPORTATION, ENVIRONMENTAL OR OTHER DOMAINS. YOU DON'T NEED TO COMMIT EARLY-TRY PROJECTS, COURSES AND SEE IF YOU ENJOY BUILDING THINGS IN THE REAL WORLD.

2.TECH

IF YOU THINK TECH IS COOL SHIT, THIS ONE'S FOR YOU. WHETHER IT'S CODING, DSA, AI, ML, SOFTWARE OR DATA, THERE'S A HUGE WORLD BEYOND YOUR BRANCH. START EXPERIMENTING, BUILD THINGS, BREAK THINGS AND LEARN. IF SOLVING PROBLEMS WITH CODE GETS YOU HOOKED,TECH MIGHT BE YOUR THING.

3.CONSULTING

LOVE TALKING TO RANDOM PEOPLE AND SOLVING RANDOM PROBLEMS? CONSULTING COULD BE YOUR FIT. FROM "HOW MANY CUPS OF AMUL ARE SOLD IN INDIA EVERY DAY?" TO TACKLING REAL BUSINESS PROBLEMS, YOU GET TO THINK FROM DIFFERENT PERSPECTIVES, WORK WITH DIFFERENT PEOPLE AND EXPLORE DIFFERENT INDUSTRIES.

4.FINANCE

IF TERMS LIKE DERIVATIVES, VALUATIONS, TRADING OR MARKETS MAKE YOU WANT TO KNOW MORE INSTEAD OF RUNNING AWAY, GIVE FINANCE A SHOT. EXPLORE HOW MONEY MOVES, HOW MARKETS WORK AND HOW BUSINESSES ARE VALUED. YOU DON'T NEED TO BE A FINANCE NERD ON DAY 1— START CURIOUS AND SEE WHERE IT TAKES YOU.

ONE LAST FUNDA: THERE IS NO "BEST" INTERNSHIP. THERE IS ONLY A DIRECTION THAT MAKES YOU WANT TO LEARN MORE. YOU HAVE TWO YEARS—USE THEM TO EXPLORE BEFORE YOU DECIDE.

The Civil Engineering

CR□SSW□RD

ACROSS

4. A rigid structural beam that extends horizontally and is supported at one end

6. Civil main bohut _____ h.

8. Crushed stone, sand, and gravel used in concrete mixtures

9. Civil engineers placement packages are known as

DOWN

1. The science of measuring land and mapping its features

2. Every first year civil engineer dream

3. A primary construction material made from cement, sand, and gravel

5. Best club of IIT Delhi

7. Popularly known by other branches

About

So... What Exactly Is CEF? Ever wondered who keeps organising the events, sending the reminders, putting up the posters, and somehow convincing you to attend? That's CEF.

The Civil Engineering Forum is the student body of the Civil Engineering Department at IIT Delhi-but we like to think we're a little more than that. We're here to make sure Civil Engineering isn't just lectures, labs, assignments, and wondering why you chose Civil. From technical sessions and workshops to Extravaganza, departmental dinners, competitions, orientations, merch, and everything in between,

CEF keeps the department busy

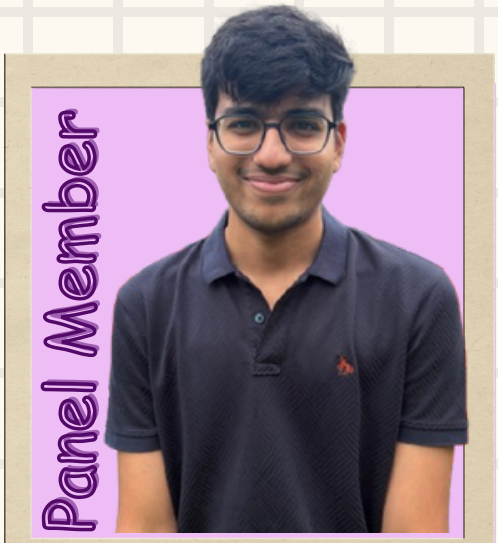
Behind it all is a bunch of students planning, designing, coordinating, improvising, and discovering that the plan has changed five minutes before the event. Or, as we like to put it:

"Bhai, aa ja, kuch toh hoga."

So come learn, participate, organise, compete, contribute or just show up for the free stuff. Welcome to CEF. We promise there's more to Civil than concrete.

The Team!!!

MEHUL



PARV

COORDINATORS

Jyoti

Manasvi

Mohit

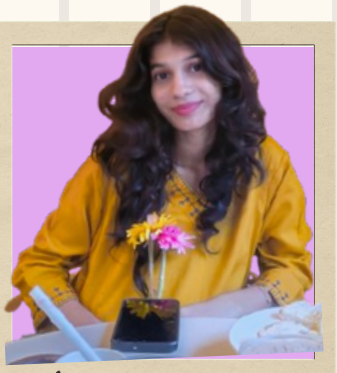
Prakhar

Prince

Kavin

Hemant

EXECUTIVES



Simran

Tanvi

Tanu

Shreyansh

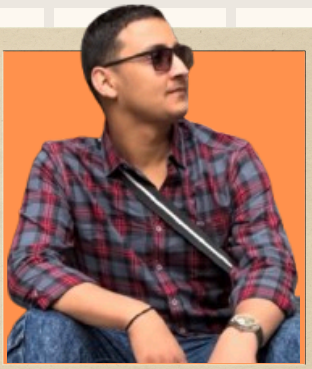
Rudra

Oshine

Sujal

Ghanshyam

Samreen



Lucky

Vridhi

Arpit

TYPES OF STUDENT

"MUJHE TO UPSC HI KARNA HAI, BAS THODA CIVIL COMPLETE KAR LOO! "- SONU

UPSC ASPIRANTS IN CIVIL

THESE STUDENTS TREAT CIVIL ENGINEERING AS JUST A STEPPING STONE. BETWEEN LECTURES AND ASSIGNMENTS, THEY'RE SECRETLY PREPARING FOR UPSC—READING NEWSPAPERS, MAKING NOTES ON POLITY, AND DAYDREAMING ABOUT BECOMING BUREAUCRATS. YOU'LL FIND THEM IN THE LIBRARY, PRETENDING TO STUDY CIVIL TOPICS WHILE ACTUALLY BRUSHING UP ON INDIAN HISTORY. THEIR CONFIDENCE IS UNMATCHED: "CIVIL TOH BAS EK STEP HAI, IAS TOH FINAL GOAL HAI!"

LAB REPORT BHEJ NA BHAI!

THE LAST-MINUTE HUSTLERS

– THESE STUDENTS THRIVE UNDER PRESSURE. THEY START THEIR PROJECTS, REPORTS OR QUIZ PREP JUST HOURS BEFORE THE DEADLINE AND YET SOMEHOW MANAGE TO SCRAPE THROUGH. ARE THEY PROCRASTINATORS OR SECRET GENIUSES? NO ONE KNOWS. WE WOULD RECOMMEND YOU NOT TO BE ONE OF THEM BUT IF YOU DO END UP IN ONE OF THESE LAST-MINUTE SITUATIONS, REMEMBER TO KEEP THAT ONE FRIEND CLOSE WHO COULD TAKE YOU OUT OF IT.

SAB HO JAAYEGA BRO, STRESS MAT LO!

THE CHILL GUYS

WHILE EVERYONE ELSE IS PULLING ALL-NIGHTERS FOR REPORTS AND QUIZZES, THESE STUDENTS WILL COME TO THE LIBRARY TO WATCH SERIES AND CHILL LIKE THEY'RE ON VACATION. THEY HAVE ACHIEVED ENLIGHTENMENT AND TRUST THE UNIVERSE TO HELP WITH EVERYTHING WITHOUT LIFTING A FINGER.

KYA KAREIN, YEH DUNIYA HI AISI HAI!

GUARDIAN ANGELS

THESE ARE THE TRUE UNSUNG HEROES OF THE BATCH—PREPARING LAB REPORTS, SHARING CLASS NOTES, AND EVEN PERFORMING ENTIRE EXPERIMENTS WHILE THE REST OF US ARE BUSY PERFECTING THE ART OF PROCRASTINATION. YOU ONLY REALIZE THEIR WORTH WHEN THE DEADLINE IS KAL-SUBAH-9-BAJE, AND PANIC SETS IN. THESE GUARDIAN ANGELS ARE THE REASON THE BATCH SURVIVES, AND YET THEIR BRILLIANCE IS NEVER OFFICIALLY ACKNOWLEDGED.

IN CIVIL DEPARTMENT

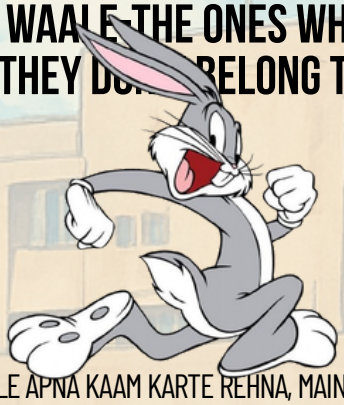
"ISS SEM TOH CHAGGI BHI NAHI AYEGI😭"

THE GASLIGHTERS

THESE PEOPLE WILL MAKE YOU QUESTION YOUR OWN SANITY. THEY WOULD APPEAR AS THE MOST DISINTERESTED PEOPLE IN THE WORLD REGARDING ACADEMICS (MOH MAYA HAI SAB BHAI), AND THEN AFTER GETTING YOUR CHECKED PAPERS, YOU WILL SEE THEM ARGUING WITH TA TO GIVE HALF NUMBER (AFTER SCORING DOUBLE THE AVERAGE). IT IS VERY PERSONAL FOR THEM TO NOT APPEAR AS A STUDIOUS PERSON (KYONKI PADHE BINA HI CG 8.5+ ATI HAI).

MAA YE MAI KAHAN AA GAYA MAA

DEPC WAALE-THE ONES WHO FEEL LIKE THEY DO[illegible]BELONG TO CIVIL



CIVIL WAALE APNA KAAM KARTE REHNA, MAIN TOH APNE REAL DEPARTMENT KI PLANNING KAR RAHA HOON!" THESE STUDENTS ARE CONVINCED THAT THEY WERE SOMEHOW MISTAKENLY PLACED IN CIVIL AND THAT THEIR "TRUE CALLING" LIES ELSEWHERE. THEY GENERALLY HANG OUT WITH PEOPLE FROM OTHER DEPARTMENTS. THEY OFTEN ACT LIKE THEY'RE "TOO GOOD" FOR CIVIL, MAKING SURE EVERYONE KNOWS THEY'LL BE MOVING TO ANOTHER DEPARTMENT SOON.

I'M IN CIVIL, BUT MY HEART BELONGS TO TECH-

CIVIL TECHIES

THESE STUDENTS ARE PHYSICALLY PRESENT IN THE CIVIL ENGINEERING DEPARTMENT BUT MENTALLY CODING, DEBUGGING, OR DREAMING OF THE NEXT TECH BREAKTHROUGH. THEY HAVE THEIR EYES ON A TECH JOB SINCE DAY ONE. THEY DON'T MISS A SINGLE CODING COMPETITION, EVEN IF IT'S DURING THE MIDSEMS AND THINK THAT SUBMITTING LAB REPORTS IS A WASTE OF TIME BECAUSE "AUTOMATION IS THE FUTURE"(WELL, THEY ARE NOT ALONE IN THIS THOUGHT).

SACCHE BRANCH-BHAKT

CORE KI OR

THOUGH THEIR NUMBER HAS DECREASED OVER THE YEARS, THEY ARE STILL FOUND IN PLENTY. THEY'RE ALWAYS THE ONES WHO DROP UNSOLICITED ADVICE: "TERA AIM HONA CHAHIYE HIGHER STUDIES, YAAR! DON'T GET STUCK IN A CORPORATE JOB, BE A PART OF THE FUTURE!" WHILE WE'RE ALL STILL FIGURING OUT LIFE POST-BTECH, THEY'RE ALREADY HALFWAY THROUGH THEIR DREAM PROGRAM, READY TO CONQUER THE WORLD OF ACADEMIA. THESE CORE SEEKERS ARE A DIFFERENT BREED—MORE FOCUSED THAN A LASER BEAM AND LESS CONCERNED ABOUT PLACEMENTS THAN YOU ARE ABOUT YOUR NEXT CUP OF COFFEE!

LEARNING BEYOND THE CLASSROOM

Field Trip >

AT CEF, LEARNING GOES BEYOND TEXTBOOKS AND CLASSROOMS. THROUGH REGULAR FIELD TRIPS AND SITE VISITS, STUDENTS GAIN PRACTICAL EXPOSURE TO REAL-WORLD PROJECTS ACROSS STRUCTURAL, GEOTECHNICAL, ENVIRONMENTAL AND OTHER AREAS OF CIVIL ENGINEERING.

THESE EXPERIENCES HELP STUDENTS CONNECT THEORY WITH PRACTICE AND UNDERSTAND HOW ENGINEERING WORKS IN THE REAL WORLD.

HOW SMALL IDEAS LEAD TO GREAT DISCOVERIES

A FLOWERY BEGINNING: HOW A GARDENER'S FLOWER POTS CHANGED CONSTRUCTION INDUSTRY FOREVER:

Joseph Monier wasn't your typical engineer. He was just a gardener with a problem: his flower pots kept breaking. Little did he know his solution would change the world of construction.

Monier loved his plants, but he hated how easily his pots cracked. So, he tried something new. He mixed cement with wire mesh to make stronger containers. It worked like magic! His new pots were tough and flexible.

In 1867, Monier patented his idea and showed it off at the Paris Exposition. But he didn't stop at flower pots. He thought, "If this works for plants, why not buildings?" Soon,

Monier was using his wire-and-cement mix (now called Reinforced Concrete, a word you will hear whenever we talk about civil engineering) for railway ties, pipes, floors, arches, and bridges.

Monier wasn't the first to think of putting metal in concrete. But he saw something others missed. He realised that concrete is excellent at handling squeezing forces (compression) while metal is excellent at dealing with stretching forces (tension), and together, they make a super-strong team!

Even though Monier didn't have formal engineering training, he understood his invention's potential. It took years for scientists to explain why it worked so well, but that didn't stop people from using it. Reinforced concrete quickly spread across Europe, becoming a cornerstone of modern construction. Today, we can thank this creative gardener for many of the buildings and bridges we see around us.

Monier's story teaches us that sometimes, the biggest breakthroughs come from unexpected places. Who knows? The next world-changing idea might be growing in your backyard!

THE WOBBLY SKYSCRAPER:
HOW A CONSTRUCTION CRISIS LED TO A BREAKTHROUGH

Imagine building a sleek, 60-story skyscraper, only to watch its windows pop out and crash to the street below. That's exactly what happened with the John Hancock Tower in Boston during the 1970s. But this problem led to an unexpected solution that changed skyscraper design forever.

The John Hancock Tower, designed by the famous architect I.M. Pei, was meant to be a gleaming addition to the Boston skyline. But there was a big problem: the wind.

On blustery days, the tower would sway back and forth. While some movement is normal for tall buildings, this was extreme.

The swaying was so bad that it made people inside feel seasick. Worse yet, it was causing the massive windows to crack and fall out!Engineers were stumped. They needed a quick fix to stop the swaying while they figured out a permanent solution. Someone came up with a bizarre idea: what if they added weight to the building?

They brought in huge water tanks - giant plastic tubes - and placed them on certain floors of the tower. The idea was simple: add mass to the building to counteract the wind's force.

To everyone's surprise, it worked! The water tanks didn't just add weight; they also sloshed around in a way that countered the building's movement. This accidental solution gave engineers a brilliant idea: what if they could design a system that did this on purpose?

This crisis-turned-discovery led to the development of the tuned mass damper (TMD). A TMD is like a giant pendulum installed at the top of a skyscraper. When the building sways one way, the TMD swings the other way, cancelling the motion.

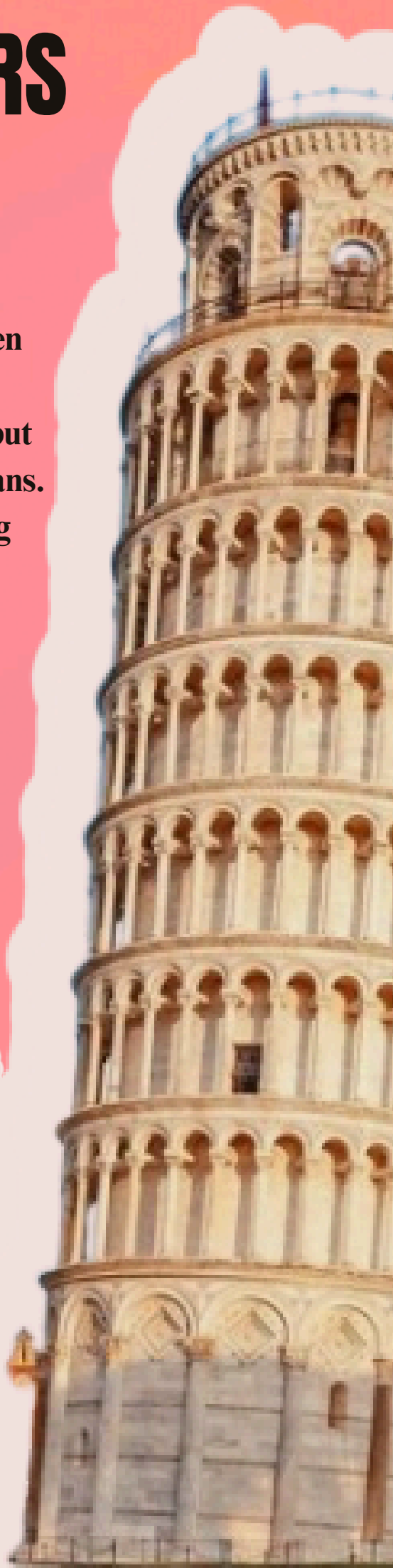
So, Next time you look at a towering skyscraper, remember: there might be a giant pendulum at the top, keeping it steady - all thanks to some water tanks and a building that wouldn't stop swaying!

CIVIL BEYOND BORDERS

The Leaning Tower of Pisa: Engineering's Happy Accident

What looks like a marvel of medieval architecture is actually a 14,500-ton lesson in what happens when the ground beneath your project won't cooperate. The Leaning Tower of Pisa wasn't meant to lean, but soft soil and a too-shallow foundation had other plans. By the time builders hit the third floor, it was tilting a disaster in slow motion.

But rather than being dismissed as a monumental failure, engineers throughout history saw a challenge. Centuries of interventions stabilized its iconic slant without ever straightening it completely. Why? Because imperfection has its charm and its tourism value. The Leaning Tower reminds us that even when things go sideways, the story you build around them can stand tall.



College Kickoff Check

Attended the wrong lecture.	Tried to decode Delhi's weather.	Joined 50 WhatsApp groups.	Made fun of civil engineering... even when you are in civil (or someone called you a mistri).
Became a pro at finding the cheapest canteen options.	Used mess food as a conversation starter.	Became a pro at resource hunting.	Tried to figure out what "DR, CR, POR, and SOP" actually mean.
Ate the Jia Sarai Poha at 5 am.	Pretended to study in the library but actually scrolled Instagram.	Participated in a hostel vs hostel sports match even if you had zero skills.	Made plans for a class trip that never happened.
Asked someone for prom.	Got a crush on a senior.	Received unwanted gyaan from seniors.	Watch the sunrise from the Main Building top floor.

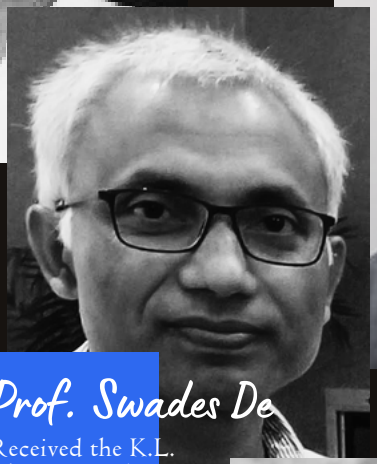
-list, Don't Miss These!

Experienced the "First Midsem Panic."	Had a deep existential crisis post-midsem exams.	Got a 0 on that first APL quiz/ Col Lab test.	Wished for longer deadlines, got them, and still missed them.
Bonded with classmates during a collective last-minute struggle for an end-sem.	Bragged to your family about doing "cutting-edge projects" while crying over tutorials.	Tried to join all the clubs and made your life a mess.	Pulled an all-nighter but still ended up ty ty fuss.
Wrote a request mail to a prof, triple-checked for typos, and got ghosted.	Tried to explain the IIT grading system to relatives and gave up.	Realized nobody takes attendance seriously—until they do.	Got into a pointless debate about which hostel is better.
Filled forms/ surveys just to get that dominos/ amazon cashback :)	Had random deep convos with bros just some hours/days before the exams	Realized that the first semester is only the trailer.	Binge-watched an entire series in one night because why not?

Faculty Awards

Vasant Matsagar

Fullbright-Kalam Climate Fellowship for Academic and Professional Excellence by the United States-India Educational Foundation (USIEF) in 2023-24.



Prof. Swades De

Received the K.L. Chopra Faculty Research Award at IIT Delhi for applied research in 2024.
Honored with the Abdul Kalam Technology Innovation National Fellowship for the period 2021-24.
Distinguished Professional, AICTE, Gov of India, 2024-27

Prof. Balaji Rajagopalan

Vaibhav Fellowship proposal accepted

Prof. Dipti Ranjan Sahoo

Election to the Indian National Academy of Engineering

Prof. Sumit Kumar Sinha

Zuhair A. Munir Award for Best Dissertation in Engineering by UC Davis in 2024.

Prof. Arnab Benerjee

Early Career Faculty Research Award, IIT Delhi

Soil Says Otherwise

you, & many others

Guys, I was talking to a few freshers today, and they have some wild myths about civil engineering.

Wanna hear them?

Oh, this gonna be good.

Let's hear it!

'Ghar ke nakshe banado'
Apparently, they think we're walking, talking AutoCAD machines who can design homes in minutes and settle family feuds over room sizes for free!

Haha! Sure, because that's exactly why we spent years studying - so we can become free family dispute consultants!

'No Scope' Can you believe it?

They think there's no scope in a field that literally builds the world -roads, bridges, skyscrapers, and even their favourite pizza joints!

But, I mean, we also thought the same at the start , isn't it?

Ironic , isn't it?

Without civil engineers, they wouldn't even have a decent hangout spot!

'Sirf aadmiyo ke liye hai'

I told them women engineers are ruling sites like pros -balancing deadlines and safety helmets while breaking stereotypes one project at a time.

True! I know plenty of women engineers who can run a site better than anyone.

This myths needs to go!

'Outdated field hai'

Outdated? We were there during the pyramids, and we'll be here when they start building colonies on Mars!

Right! The only thing older than civil engineering is the ground itself.

'Bas cement hi toh milana hai'

Tell them if it was that easy, DIY YouTube builders would be creating their own Taj Mahals. They don't realize it's science, math, and countless calcuations.

Haha, true! We make calculators and even software sweat for real.

'JCB ke best friends'

This one was funny -they think we're best friends with JCBs!

Well, to be fair, who wouldn't want a giant yellow machine as a best friend? JCBs don't argue, they just dig in!

'Civil engineer aur architect same hi toh hai'

They actually think civil engineers and architects are the same! I told them that's like saying doctors and pharmacists are the same.

Perfect example! Architects design, we make sure it stands and doesn't collapse. But what can we expect from them even our parents and relatives also think the same :((

These myths are hilarious!

Freshers really need some serious myth-busting sessions.

Totally! Maybe we should run a 'Mythbusters: Civil Engineering Edition' during orientation week.

I'm in!

Let's bust myths and shows them what real civil engineering is all about.

@CEF.IITD